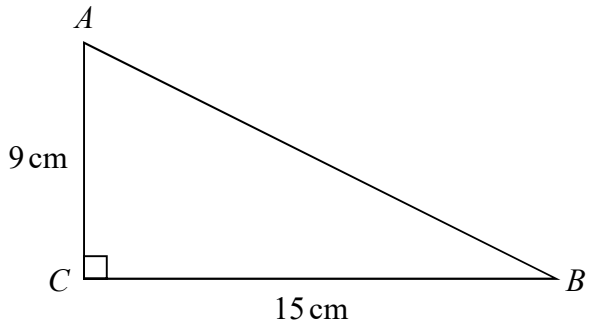
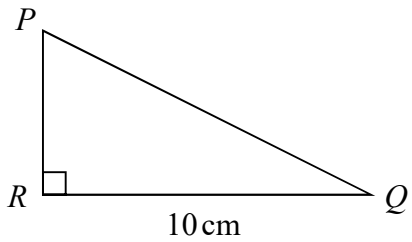
29 *ABC* and *PQR* are similar right-angled triangles.





angle *ABC* = angle *PQR*

(a) Work out the length of *PR*.

BC corresponds to RQ: scale = 10/15 = 2/3

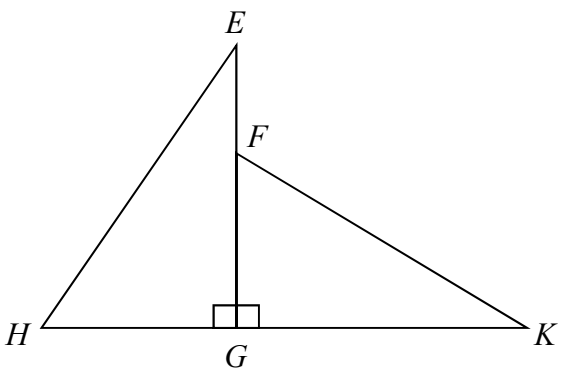
AC corresponds to PR: PR = 9 × 2/3 = 6 cm

2 marks total
Match corresponding sides
PR = 6 cm

..... cm

(2)

Triangle *EGH* is congruent to triangle *KGF*.



HK = 10 cm.
HG = 4 cm.

(b) Work out the length of *EF*.

GK = HK − HG = 10 − 4 = 6 cm

By congruence EG = GK = 6 cm; FG = HG = 4 cm

EF = EG − FG = 6 − 4 = 2 cm

2 marks total
Use congruent sides
EF = 2 cm

..... cm

(2)

(Total for Question 29 is 4 marks)